Swinburne University of Technology

COS20007 Object Oriented Programming

2.4P - Case Study Iteration 1 - Identifiable Object

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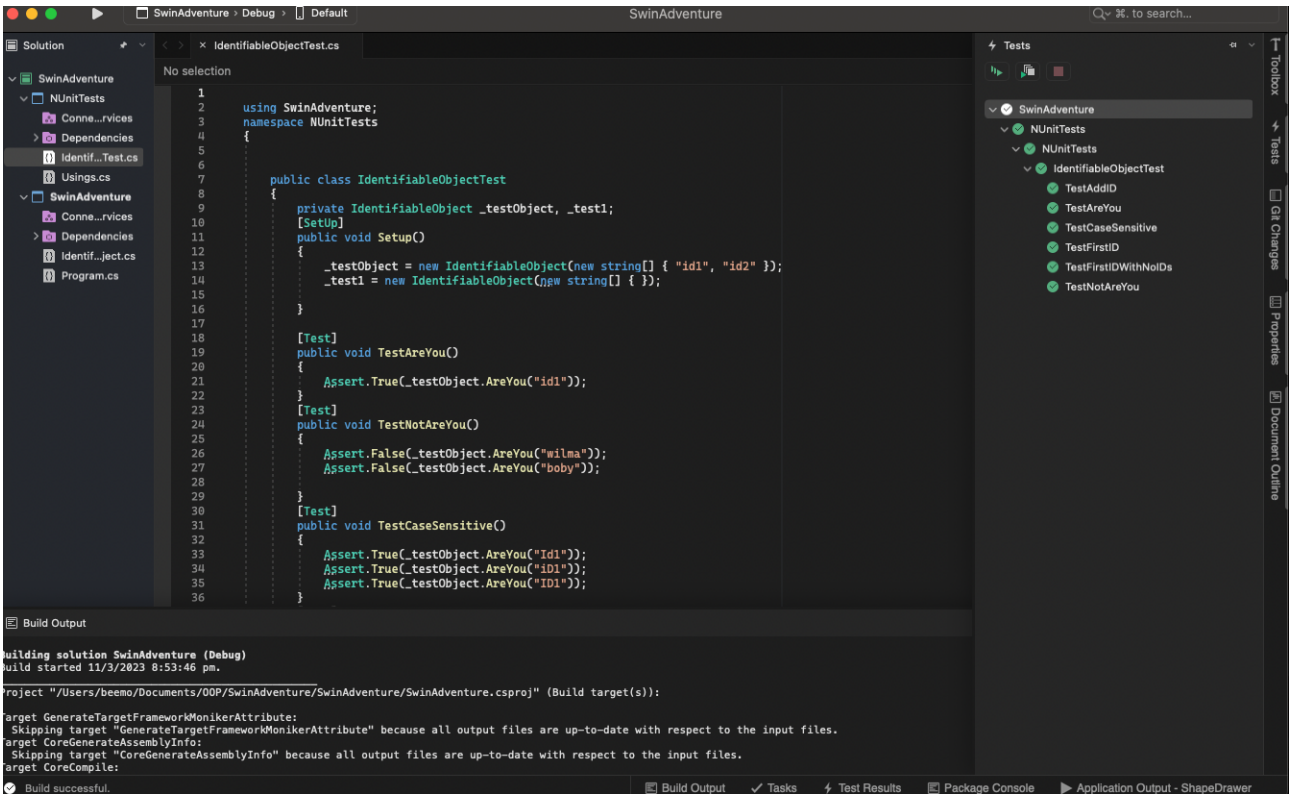
```
using System;

namespace SwinAdventure
{
    public class IdentifiableObject
    {
        private List<string> _identifiers = new List<string>();

        public string FirstId
        {
            get
            {
                if (_identifiers.Count == 0)
                    return "";
                return _identifiers[0];
            }
        }



        public IdentifiableObject(string[] idents) //create constructor
        {
            _identifiers.AddRange(idents);
        }
        public bool AreYou(string id)
        {
            id = id.ToLower();
            foreach (string idents in _identifiers)
            {
                if (idents == id)
                {
                    return true;
                }
            }
            return false;

        }
        public void AddIdentifier(string id)
        {
            id = id.ToLower();
            _identifiers.Add(id);

        }

    }
}

```

```

using SwinAdventure;
namespace NUnitTests
{


    public class IdentifiableObjectTest
    {
        private IdentifiableObject _testObject, _test1;
        [SetUp]
        public void Setup()
        {
            _testObject = new IdentifiableObject(new string[] { "id1", "id2" });
            _test1 = new IdentifiableObject(new string[] { });

        }

        [Test]
        public void TestAreYou()
        {
            Assert.True(_testObject.AreYou("id1"));
        }
        [Test]
        public void TestNotAreYou()
        {
            Assert.False(_testObject.AreYou("wilma"));
            Assert.False(_testObject.AreYou("boby"));

        }
        [Test]
        public void TestCaseSensitive()
        {
            Assert.True(_testObject.AreYou("Id1"));
            Assert.True(_testObject.AreYou("iD1"));
            Assert.True(_testObject.AreYou("ID1"));
        }
        [Test]
        public void TestFirstID()
        {
            Assert.That(_testObject.FirstId, Is.EqualTo("id1"));
        }
        [Test]
        public void TestFirstIDWithNoIDs()
        {
            Assert.That(_test1.FirstId, Is.EqualTo(""));
        }
        [Test]
        public void TestAddID()
        {
            _testObject.AddIdentifier("id3");
            Assert.True(_testObject.AreYou("id1"));
            Assert.True(_testObject.AreYou("id2"));
            Assert.True(_testObject.AreYou("id3"));
```

```

        }
    }
}
```


SwinAdventure › Debug › Default
SwinAdventure
Solution
SwinAdventure
NUnitTests
Conne...rvices
Dependencies
Identif...Test.cs
Usings.cs
SwinAdventure
Conne...rvices
Dependencies
Identif...ject.cs
Program.cs
IdentifiableObjectTest.cs
No selection
using SwinAdventure;
namespace NUnitTests
{
public class IdentifiableObjectTest
{
private IdentifiableObject _testObject, _test1;
[SetUp]
public void Setup()
{
_testObject = new IdentifiableObject(new string[] { "id1", "id2" });
_test1 = new IdentifiableObject(new string[] { });
}
[Test]
public void TestAreYou()
{
Assert.True(_testObject.AreYou("id1"));
}
[Test]
public void TestNotAreYou()
{
Assert.False(_testObject.AreYou("wilma"));
Assert.False(_testObject.AreYou("boby"));
}
[Test]
public void TestCaseSensitive()
{
Assert.True(_testObject.AreYou("Id1"));
Assert.True(_testObject.AreYou("iD1"));
Assert.True(_testObject.AreYou("ID1"));
}
Tests
SwinAdventure
NUnitTests
NUnitTests
IdentifiableObjectTest
TestAddID
TestAreYou
TestCaseSensitive
TestFirstID
TestFirstIDWithNoIDs
TestNotAreYou
Build Output
uilding solution SwinAdventure (Debug)
uild started 11/3/2023 8:53:46 pm.
roject "/Users/beemo/Documents/OOP/SwinAdventure/SwinAdventure/SwinAdventure.csproj" (Build target(s)):
arget GenerateTargetFrameworkMonikerAttribute:
Skipping target "GenerateTargetFrameworkMonikerAttribute" because all output files are up-to-date with respect to the input files.
arget CoreGenerateAssemblyInfo:
Skipping target "CoreGenerateAssemblyInfo" because all output files are up-to-date with respect to the input files.
arget CoreCompile:
Build successful.
Build Output
Tasks
Test Results
Package Console
Application Output - ShapeDrawer